Can Frozen Text and Code Embeddings Screen Coding-Agent Failures?

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Abstract

Release engineers need to screen completed coding-agent runs before integrating candidate patches into production repositories. We study task-end anomaly screening (**Category 4: Text / Logs**) on 284 official `SWE-agent` (`GPT-4`) trajectories (54 resolved normal, 230 unresolved failure; 80.99% failure prevalence) using local, frozen `all-MiniLM-L6-v2` log embeddings and `codebert-base` patch diff embeddings, and validate once on an untouched external holdout of 271 `SWE-agent` (`Claude-3-Opus`) runs (35 normal, 236 failure). Under 5-fold stratified evaluation calibrated at validation FPR $\leq 20\%$, our primary Tier 2 dual-encoder k-NN detector reaches only 0.481 pooled out-of-fold (OOF) balanced accuracy (BA $= 0.479 \pm 0.05$ across folds), flagging $14/54$ successes (FPR $= 0.259$) while missing $179/230$ failures (recall $= 0.222$), vs. Tier 1 rule heuristics (BA $= 0.526$) and Tier 3 supervised encoder logistic regression (BA $= 0.596$, $p = 0.003$). Single-factor ablations reveal why equal-weight multimodal fusion fails: noisy `MiniLM` log distances exhibit $56\times$ larger variance than `CodeBERT` patch distances, contributing 98.24% of unweighted distance variance and drowning out the informative patch channel (`CodeBERT` patch-only k-NN reaches BA $= 0.594$, ΔBA $= +0.113$, $p = 0.012$). Meanwhile, parameter-free `Inductive ECOD` on 23 cleaned workflow/patch statistics reaches BA $= 0.657$ (FPR $= 0.130$, $p = 0.004$) and a task-end `Conditional Residual` detector reaches BA $= 0.675$ ($p = 0.011$). Zero-shot transfer and 5-fold replication on the untouched `Claude-3-Opus` holdout preserve the full ranking (T2 $= 0.542/0.536$, `CodeBERT` patch-only $= 0.620/0.592$, `ECOD` $= 0.645/0.645$, `Conditional Residual` $= 0.672/0.708$, T3 $= 0.691/0.711$). Finally, an operational cost model ($1,200 per escaped failure vs. $30 per reviewed alert; break-even prevalence $\pi^* = 2.5\%$) shows that at 81% failure prevalence, universal review ($15,000) dominates automated gates ($23,028–$381,602), whereas anomaly detectors provide unique value on the rare 6.0% $(17/284)$ *silent semantic failure* subtype where surface rules catch $0/17$ (0.0%) and `Conditional Residual` catches $8/17$ (47.1%).

1 Introduction

Autonomous coding agents iteratively issue shell commands, inspect tracebacks, and edit files to resolve software issues (Yang et al., 2024), yet completed runs frequently fail regression suites (Jimenez et al., 2024). Our stakeholder is a **release engineer** triaging completed agent runs before running costly full-repository integration suites or merging diffs: an alert routes a patch to mandatory human review, whereas unflagged runs proceed to standard CI testing (never autonomous release). We formulate screening at **task-end** rather than prefix stopping because release engineers receive candidate diffs only at trajectory completion, and prefix aborts prematurely terminate recoverable debugging loops (e.g., resolved run `scikit-learn-14894` executes 25 steps with 19 error observations before converging on a valid fix).

Our study falls under **Category 4: Text / Logs**. Agent logs mix shell commands, tracebacks, and unified git diffs. Scalar counts discard code semantics, yet pre-trained encoders measure topical similarity rather than functional correctness. We ask: *Can local, frozen text and code embeddings improve task-end failure screening over clean numeric workflow statistics under a shared normal-reference protocol and false-alarm constraint?*

We model resolved runs ($y = 0$, normal reference) as our baseline and unresolved runs ($y = 1$, failure) as the target event. Surface counts poorly separate outcomes: resolved `django-13230` takes 15 steps with 9 error observations and is falsely flagged by our primary text/code detector ($s = 0.1128 > \tau = 0.1087$), whereas failed `sympy-12454` finishes in 11 steps with 1 error observation and escapes ($s = 0.0248 < \tau = 0.0731$). Crucially, we address a core domain tension: across the benchmark,

failures constitute 80.99% (230/284) of generated runs. Modeling minority successes (19.01%, 54/284) as the normal reference class is required because valid repairs share a coherent workflow-to-patch manifold, whereas failures fail heterogeneously. To connect majority-failure screening to classical rare-anomaly detection, we also isolate the rare **silent semantic failure subtype** (6.0%, 17/284): runs submitting a non-empty patch with `last_step_error=0` and $\leq$ 1 error step that nonetheless fail unit tests. Our contributions are: (1) a shared-split 3-tier benchmark across four representations (R0–R3) with paired t-tests, bootstrap CIs, and latencies (Table 2); (2) mechanistic variance and neighbor-hit proofs closing three hypotheses (Tables 3–4); (3) once-only confirmatory evaluation on 271 untouched `Claude-3-Opus` runs; and (4) cost-aligned threshold calibration ($\pi^* = 2.5\%$, Table 5) and step/feature error attributions (Table 6, Figure 5).

2 Related Work

`SWE-bench` (Jimenez et al., 2024) evaluates patch generation on 2,294 GitHub issues across 12 Python repositories (best original baseline: 1.96% resolved; 4.33% on 300-task `Lite`). `SWE-agent` (Yang et al., 2024) introduces an agent–computer interface achieving 12.5% pass@1 on full `SWE-bench` (18.0% on `Lite`) and 87.7% on `HumanEvalFix`. Whereas those works evaluate issue resolution via container test suites, we study **Category 4: Text / Logs** trajectory screening prior to test execution. Sentence-BERT (Reimers and Gurevych, 2019) learns Siamese transformer embeddings for cosine similarity; we use frozen `all-MiniLM-L6-v2` (384-D) with chunk pooling over logs. `CodeBERT` (Feng et al., 2020) pre-trains a 12-layer bimodal RoBERTa model (768-D) on code and text; we adapt frozen `codebert-base` to embed diff hunks without task fine-tuning. Among anomaly detectors, Deep SVDD (Ruff et al., 2018) fits a neural hypersphere to normal data but risks hypersphere collapse when $n_{\text{ref}} \approx 32$. We instead pair non-parametric cosine k-NN in frozen embedding space (Tier 2) with tabular `Isolation Forest` (Liu et al., 2008) and an **inductive** adaptation of parameter-free `ECOD` (Li et al., 2022) that freezes empirical tail CDFs on normal training folds, alongside Tier 3 supervised logistic regression and `XGBoost` (Chen and Guestrin, 2016).

3 Task Definition and Data

Primary cohort and untouched confirmatory holdout. Our primary dataset comprises all $N = 284$ generated trajectories from the official `20240402_sweagent_gpt4` `SWE-bench Lite` run[1] (16 `no_generation` aborts excluded a priori): $n_0 = 54$ resolved ($y = 0$, normal reference, 19.01%) and $n_1 = 230$ unresolved ($y = 1$, failure, 80.99%). Because the `gpt4` cohort was inspected in earlier debugging drafts, repartitioning alone cannot establish confirmatory validity. We therefore acquired an untouched external cohort: all $N = 271$ generated runs from `20240402_sweagent_claude3opus` (29 `no_generation` runs excluded; $n_0 = 35$ normal, 12.92%; $n_1 = 236$ failure, 87.08%). SHA-256 hashes and official labels are archived in `Dataset/provenance.json` and `Dataset/v2/claude3opus_provenance.json`. We retain natural prevalence (80.99%, baseline AP $= 0.810$) and additionally track the rare 6.0% (17/284) **silent semantic failure** subtype (`last_step_error=0`, error_count ≤ 1, non-empty patch, $y = 1$).

Leakage hygiene and representations (R0–R3). Feature extraction uses only `action`, `observation`, and final `submission` diffs; `thought/response` text, gold patches, evaluator logs, exit codes, and instance/repo IDs are stripped. Normalization strips ANSI codes, hex addresses, and `<WORKDIR>` prefixes while keeping relative paths and traceback tokens (verified via full-cohort hash audits and 20 manual trajectory inspections). We compare four representations: (1) **R0 / R0-clean (24 / 23 tabular features):** 16 workflow counts (steps, errors, streaks, stalled turns, tokens, cost) and 8 patch metrics (chars, lines added/removed, files, hunks, edit ratio, test flag). `R0-clean` drops `total_commands` (identical to `n_steps`, $\rho = 1.000$), applies $\log(1 + x)$ to 16 skewed counts, keeps 4 binary flags, and robust-scales continuous columns via median/IQR fit *strictly on* $y = 0$ *training rows*; (2) **R1:** char 3–5-gram TF-IDF (sublinear TF, min_df $= 2$, $\leq$ 20,000 features/channel) fit on training folds; (3) **R2:** 384-D `MiniLM` log embeddings; and (4) **R3:** dual 384-D `MiniLM` log (l_x) + 768-D `CodeBERT` diff (p_x) embeddings.

4 Exploratory Feature Analysis

Restricted strictly to Fold-0 training data ($n = 170$: 32 normal, 138 failure), Table 1 and Figure 1 show that failures consume roughly 2$\times$ more received tokens (2,934 vs. 1,377, $W = 0.999$, $p = 5.5 \times 10^{-7}$) and steps (24 vs. 12, $W = 0.885$) and produce larger diffs (17 vs. 4 lines added, $D_{\text{KS}} =$

[1] https://github.com/SWE-bench/experiments/tree/main/evaluation/lite/20240402_sweagent_gpt4

Feature	Norm. (IQR)	Fail. (IQR)	W	D_{KS}	p-val
Received tokens	1377 (1042)	2934 (2658)	0.999	0.520	5.5e-07
API cost (USD)	1.24 (0.91)	2.56 (2.64)	0.892	0.440	4.5e-05
Trajectory steps	12 (8)	24 (18)	0.885	0.426	9.1e-05
Error observations	3 (3)	7 (10)	0.709	0.408	2.1e-04
Patch chars	822 (500)	1372 (1321)	0.203	0.385	5.7e-04
Patch lines added	4 (6)	17 (28)	0.178	0.554	6.3e-08

Table 1: Fold-0 training-only ($n = 170$: 32 normal, 138 failure) medians (IQR), standardized Wasserstein distance W, and two-sample KS statistics (D_{KS}, p-value).

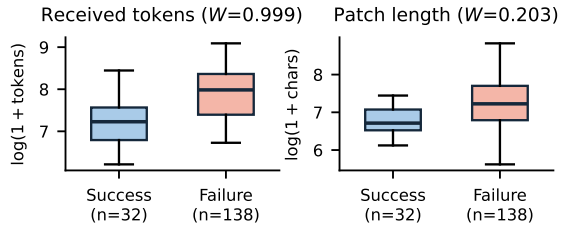


Figure 1: Fold-0 training-only $\log(1+x)$ boxplots for received tokens ($W = 0.999$) and patch length ($W = 0.203$).

0.554, $p = 6.3 \times 10^{-8}$), but exhibit heavy tails, quartile overlap, and high collinearity ($\rho = 0.985$ between tokens and cost; $\rho = 0.941$ between steps and observations). This motivates three falsifiable hypotheses: **H1 (Representation):** dual frozen text/code embeddings (`R3`) improve k-NN BA over `R0-clean` numeric and `R1` TF-IDF inputs; **H2 (Channel attribution):** `CodeBERT` patch embeddings add discriminative signal beyond `MiniLM` logs (`R2`); and **H3 (AD contamination):** replacing 5% $(1/32)$ and 10% $(3/32)$ of normal training references with failures degrades `T2` and `Inductive ECOD` BA.

5 Method

Six-checkpoint primary pipeline (Tier 2: `knn_R3_T2`). Figure 2 depicts our primary pipeline: (1) **Inputs & Hygiene** normalizes permitted action/observation logs and diffs; (2) **Frozen Encoding** chunks each step into 224-token windows with 32-token overlap (within `all-MiniLM-L6-v2`'s 256-token limit), mean-pools non-special tokens across chunks and steps, and ℓ_2-normalizes to $l_x \in \mathbb{R}^{384}$; unified diff hunks are chunked into 480-token windows with 64-token overlap (within `codebert-base`'s 512-token limit), token-weight pooled, and ℓ_2-normalized to $p_x \in \mathbb{R}^{768}$; (3) **Normal Reference Index** stores only resolved training runs $\mathcal{R}_{\mathrm{norm}}$ ($y = 0$, $n_{\mathrm{ref}} \in \{32, 33\}$); (4) **Dual k-NN Score** computes channel cosine distances $d_{\log}(x,r) = \frac{1-\cos(l_x,l_r)}{2}$ and $d_{\mathrm{patch}}(x,r) = \frac{1-\cos(p_x,p_r)}{2}$ (0 if both empty, 1 if one empty), fuses them via unweighted average

$$d(x,r) = \tfrac{1}{2}\big(d_{\log}(x,r) + d_{\mathrm{patch}}(x,r)\big) \in [0,1], \tag{1}$$

and scores $s(x) = \frac{1}{k}\sum_{r \in \mathcal{N}_k(x;\mathcal{R}_{\mathrm{norm}})} d(x,r)$ for $k \in \{1,3,5\}$ ($\leq 15.6\%$ of $n_{\mathrm{ref}} = 32$ to preserve locality); (5) **Validation Calibration** selects (k^*, τ^*) on validation to maximize recall subject to validation FPR $\leq 20\%$ (ties broken by BA, higher τ, smaller k); and (6) **Triage & Output** freezes (k^*, τ^*) to flag test runs with $s(x) \geq \tau^*$.

Baselines and sample-size hyperparameter justification. Every hyperparameter is bounded by $n_{\mathrm{ref}} \approx 32$ normal fitting rows ($n_{\mathrm{fit}} \approx 170$ total): **Tier 1 (`heuristic_T1`)** scores $\mathrm{error_rate} + 0.1\,\mathrm{repeated_tool_streak} + \mathbb{I}[\mathrm{last_step_error}] + 0.5\,(1 - \mathbb{I}[\mathrm{repro_command}])$. **Tier 2 Tabular Normal-Only Controls:** `Inductive ECOD` (Li et al., 2022) has *zero* tuned hyperparameters, freezing smoothed empirical tail CDFs $(1 + \#\{r_j \leq x_j\})/(n_{\mathrm{ref}} + 1)$ on $\mathcal{R}_{\mathrm{norm}}$ across all 23 `R0-clean` features. `Isolation Forest` (Liu et al., 2008) uses 200 trees with sub-sampling capped at 32 to bound split variance. `Numeric AE` (`24-16-8-16-24` on `R0`) uses 100 full-batch Adam epochs ($\mathrm{lr} = 10^{-3}$), eliminating mini-batch noise on 32 rows. **Tier 2 Secondary Task-End Models:** `Conditional Residual` fits ridge regression ($\alpha = 10.0$, strong ℓ_2 shrinkage for $p = 16$ behavior predictors on $n_{\mathrm{ref}} = 32$ rows, $p/n = 0.5$) and ℓ_2 logistic regression ($C = 0.1$) to predict patch features from workflow features on $\mathcal{R}_{\mathrm{norm}}$, scoring standardized output residuals. `Denoising AE` (`23-16-4-16-23` on `R0-clean`) uses 10% masking, Huber ($\delta = 1.0$) + BCE loss, and 120 full-batch AdamW epochs ($\mathrm{lr} = 10^{-3}$, weight decay 10^{-4}). **Tier 3 Supervised References:** `T3` (`R3`) and `R1` (TF-IDF) fit balanced-class ℓ_2 logistic regression ($C \in \{0.01, 0.1, 1, 10\}$); `XGBoost` (Chen and Guestrin, 2016) uses `max_depth=3` (≤ 8 leaves to prevent memorizing 170 rows), 100 trees, $\mathrm{lr} = 0.05$, and 0.8 row/column subsampling.

6 Experimental Setup

All models share an identical 5-fold stratified outer split (`seed=42`) on the 284 `GPT-4` runs, with inner stratified splits (`seed=100+f`) yielding 60% fitting ($n_{\mathrm{fit}} \approx 170$, $n_{\mathrm{ref}} \in \{32, 33\}$ normal) and 20% validation ($n_{\mathrm{val}} \approx 57$, 10–11 normal). Normal-only models fit strictly on $y = 0$ fitting rows; validation rows select hyperparameters and τ^* without refitting. Stochastic models run across seeds `42, 123, 456`. On the untouched 271-run `Claude-3-Opus` external holdout, we run once under two locked protocols: **Mode A (Zero-Shot Transfer, ZS)** ap-

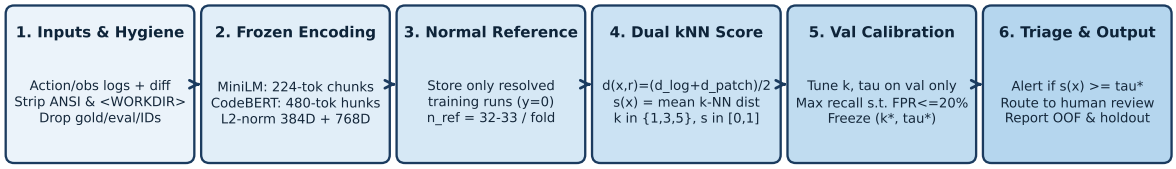


Figure 2: Six-checkpoint end-to-end architecture for our primary Tier 2 normal-reference dual-encoder k-NN pipeline (`knn_R3_T2`). All encoding, reference indexing, and threshold calibration occur strictly inside each fold.

Method	Pool BA	Fold BA±SD	FPR	Recall	TN/FP/FN/TP	Fold AP±SD	p-val	Train (s)	ms/run	Holdout BA (ZS / CV)
Always alert (context)	0.500	0.500±0.00	1.000	1.000	0/54/0/230	0.810±0.01	–	–	–	–
Never alert (context)	0.500	0.500±0.00	0.000	0.000	54/0/230/0	0.810±0.01	–	–	–	–
T1: Rule heuristic	0.526	0.526±0.09	0.204	0.257	43/11/171/59	0.864±0.05	0.310	0.00	0.000	0.501 / 0.508
T2: Dual encoder kNN	0.481	0.479±0.05	0.259	0.222	40/14/179/51	0.841±0.02	–	0.69+3151	0.000	0.542 / 0.536
T3: Encoder logistic	0.596	0.594±0.02	0.352	0.543	35/19/105/125	0.905±0.02	0.003	0.00+3151	0.018	0.691 / 0.711
R1: TF-IDF logistic	0.652	0.652±0.05	0.352	**0.657**	35/19/79/151	**0.923**±0.03	0.006	92.06	34.410	–
Inductive ECOD (R0-clean)	0.657	0.654±0.05	**0.130**	0.443	47/7/128/102	0.913±0.03	0.004	0.01	0.003	0.645 / 0.645
Isolation Forest (R0-clean)	0.635	0.632±0.02 (0.015)	0.148	0.417	46/8/134/96	0.914±0.03	0.004	0.26	0.240	0.618 / 0.595
Conditional residual (sec.)	**0.675**	0.671±0.07	0.185	0.535	44/10/107/123	0.908±0.04	0.011	0.01	0.019	0.672 / 0.708
Denoising AE (sec.)	0.648	0.646±0.05 (0.018)	0.148	0.443	46/8/128/102	0.909±0.03	0.020	0.02	2.233	–
Numeric AE (v1 control)	0.651	0.647±0.06 (0.034)	0.185	0.487	44/10/118/112	0.904±0.05	0.002	0.01	0.649	–
XGBoost (v1 control)	0.646	0.643±0.05	0.278	0.570	39/15/99/131	0.908±0.04	0.008	0.02	0.304	0.593 / 0.693

Table 2: Out-of-fold (OOF) operating points over all 284 `GPT-4` trajectories (54 normal, 230 failure) under validation FPR $\leq 20\%$ calibration. **Fold BA±SD**: 5-fold mean±SD (3-seed SD in parentheses for stochastic models). p**-val**: paired t-test across 5 folds vs. `T2`. **Train (s) / ms/run**: per-fold fit time and warmed inference latency (+3151 ms for frozen CPU encoder extraction). **Holdout BA (ZS / CV)**: untouched 271-run `Claude-3-Opus` external holdout BA in Mode A zero-shot cross-agent transfer (**ZS**) and Mode B 5-fold confirmatory replication (**CV**).

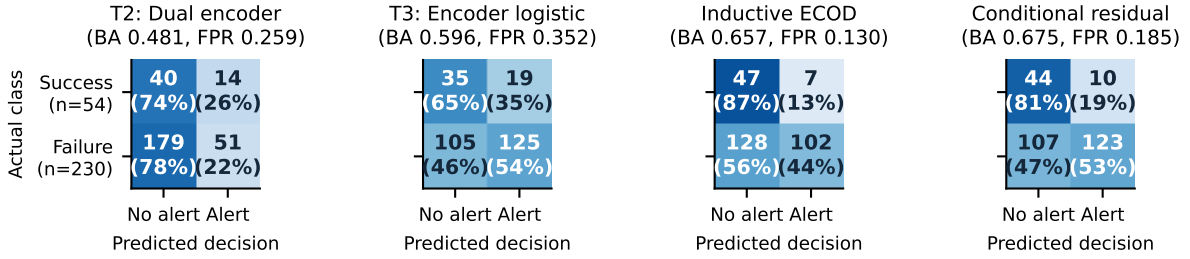


Figure 3: Pooled OOF confusion matrices across 284 `GPT-4` runs ($n_0 = 54, n_1 = 230$), colored by row-normalized class percentage. Primary `T2` leaves 219 runs unflagged (40 successes, 179 missed failures; 18.3% success precision).

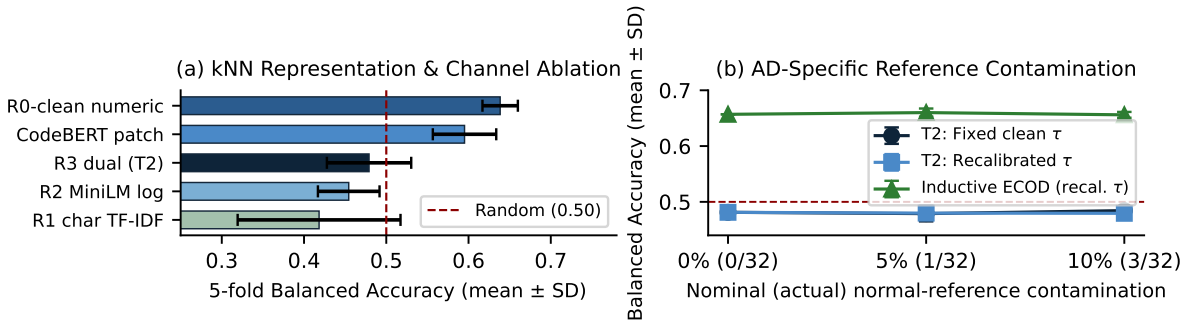


Figure 4: Single-factor ablations: (a) k-NN representation/channel ablation (5-fold mean±SD), showing CodeBERT patch-only (0.595 ± 0.04) and `R0-clean` (0.638 ± 0.02) beating unweighted dual fusion (0.479 ± 0.05); (b) AD reference contamination across 0%, 5% (1/32), and 10% (3/32) injected failures for `T2` and `Inductive ECOD`.

plies the 5 frozen `GPT-4` fold models and frozen cutoffs τ^* via majority vote, and **Mode B (Confirmatory CV)** executes the locked 5-fold pipeline ($n_{\text{ref}} = 21$ normal fitting runs/fold) without retuning. Because τ_f^* varies across folds, we report **mean within-fold AP** (Fold AP ± SD) alongside pooled and 5-fold BA, FPR, Recall, raw confusion counts, 5-fold paired t-tests vs. `T2`, and 2,000-resample stratified paired bootstrap 95% CIs.

7 Results

Main three-tier comparison and confirmatory holdout. Table 2 and Figure 3 show that primary Tier 2 dual-encoder k-NN (`T2`) reaches only BA = 0.481 (0.479 ± 0.05), FPR = 0.259

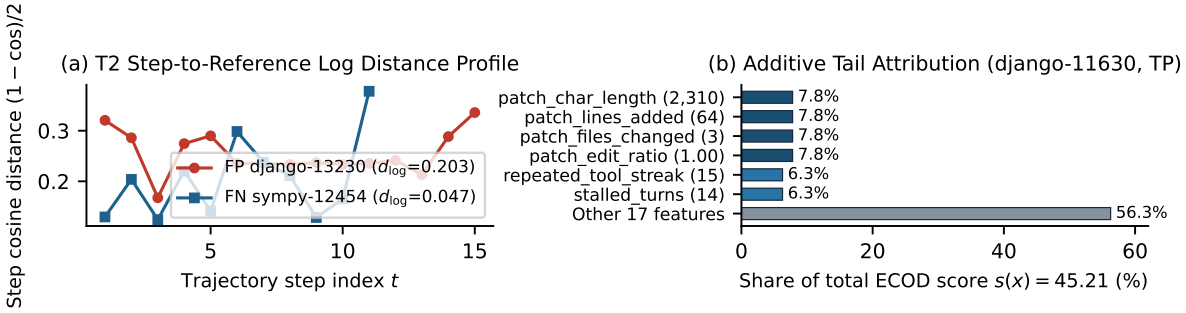


Figure 5: Interpretability and error diagnosis: (a) Step-level `MiniLM` distance $\frac{1-\cos}{2}$ to nearest normal reference for FP `django-13230` ($d_{\log} = 0.203$, noisy debugging before a valid 2-line fix) vs. FN `sympy-12454` ($d_{\log} = 0.047$, quiet run with incomplete logic); (b) Additive per-feature tail attribution for `Inductive ECOD` on TP `django-11630` ($s(x) = 45.21 > \tau^* = 33.42$).

k-NN Channel	BA (Fold±SD)	ΔBA [95% CI]	p	FPR/Rec.
R0-clean num.	**0.641** (0.64±0.02)	+0.159 [+0.09,+0.23]	0.005	0.17/0.45
R1 char TF-IDF	0.419 (0.42±0.10)	-0.062 [-0.15,+0.03]	0.386	0.30/0.13
R2 MiniLM log	0.456 (0.45±0.04)	-0.025 [-0.06,+0.00]	0.160	0.30/0.21
CodeBERT patch	0.594 (0.60±0.04)	+0.113 [+0.03,+0.20]	0.012	0.19/0.37
R3 dual (T2)	0.481 (0.48±0.05)	–	–	0.26/0.22

Table 3: Representation and channel ablation holding k-NN and validation FPR $\leq 20\%$ fixed (ΔBA, 95% bootstrap CI, and paired t-test p-value vs. `R3 dual T2`).

Detector (τ mode)	Contam. r/n_{ref}	BA±SD	ΔBA	FPR/Rec.
T2 (Clean τ)	0/32 (0%)	0.481±0.000	+0.000	0.26/0.22
T2 (Recal. τ)	0/32 (0%)	0.481±0.000	+0.000	0.26/0.22
T2 (Clean τ)	1/32 (3.1%)	0.479±0.014	-0.002	0.27/0.22
T2 (Recal. τ)	1/32 (3.1%)	0.480±0.011	-0.001	0.26/0.22
T2 (Clean τ)	3/32 (9.4%)	0.484±0.007	+0.003	0.25/0.22
T2 (Recal. τ)	3/32 (9.4%)	0.479±0.002	-0.002	0.25/0.21
ECOD (Recal. τ)	0/32 (0%)	0.657±0.000	+0.000	0.13/0.44
ECOD (Recal. τ)	1/32 (3.1%)	0.660±0.007	+0.003	0.12/0.44
ECOD (Recal. τ)	3/32 (9.4%)	0.656±0.005	-0.001	0.14/0.45

Table 4: AD-specific normal-reference contamination across 3 seeds (`42, 123, 456`) at fixed $n_{\text{ref}} \in \{32, 33\}$.

(14/54), and Recall = 0.222 (51/230, missing 179 failures). Tier 1 reaches BA = 0.526 ($p = 0.310$), whereas Tier 3 supervised encoder logistic regression reaches BA = 0.596 ($p = 0.003$) and `R1` TF-IDF reaches 0.652 ($p = 0.006$, Fold AP = 0.923 ± 0.03). Crucially, normal-only tabular detectors on `R0-clean` beat all text/code models: `Inductive ECOD` reaches BA = 0.657 (ΔBA = +0.176, 95% bootstrap CI $[+0.114, +0.240]$, paired t-test $p = 0.004$) with FPR = 0.130 (7/54, ΔFPR = −0.130, CI $[-0.241, -0.037]$), and `Conditional Residual` achieves the best overall BA = 0.675 (FPR = 0.185, Recall = 0.535, $p = 0.011$) at 0.003–0.019 ms/run vs. 3,151 ms/run for CPU transformer extraction. On the untouched 271-run `Claude-3-Opus` holdout (Table 2, last column), both Mode A (ZS) and Mode B (CV) replicate the exact hierarchy: `Conditional Residual` (0.672/0.708, $p = 0.013$), `T3` (0.691/0.711, $p = 0.013$), and `Inductive ECOD` (0.645/0.645, $p = 0.068$) strongly beat `T2` (0.542/0.536) and `T1` (0.501/0.508).

Model (Val Rule)	$\bar{\tau}^*$	BA	FPR/Rec.	F_2	Cost ($)
T2 (FPR≤20%)	0.091	0.481	0.26/0.22	0.259	$381,602
T2 (Max val BA)	0.070	0.524	0.57/0.62	0.654	$192,993
T2 (Max val F_1)	0.050	0.541	0.85/0.93	0.910	$45,475
T2 (Max F_2/Min$)	0.046	0.515	0.93/0.96	0.924	$35,387
ECOD (FPR≤20%)	33.43	0.657	0.13/0.44	0.496	$276,180
ECOD (Max val BA)	31.08	0.646	0.26/0.55	0.598	$225,053
ECOD (Max val F_1)	17.71	0.540	0.89/0.97	0.936	$29,102
ECOD (Max F_2/Min$)	17.05	0.528	0.93/0.98	0.945	$23,028
Always alert (Bayes)	$-\infty$	0.500	1.00/**1.00**	**0.955**	**$15,000**

Table 5: AD threshold calibration and dollar cost (scaled to 500 runs: $30/review, $1,200/escaped FN; break-even $\pi^* = 2.5\%$).

Ablation 1: Representation & channel variance decomposition (H1 & H2). Table 3 and Figure 4(a) hold k-NN and validation calibration fixed. **H1 is refuted:** `R0-clean` numeric k-NN reaches BA = 0.641 (ΔBA = +0.159, $p = 0.005$), beating all text/code k-NN variants. **H2 is supported for patch utility, but equal-weight fusion fails:** `CodeBERT` patch-only k-NN reaches BA = 0.594 (ΔBA = +0.113 over `R3`, $p = 0.012$; +0.138 over `R2` log-only 0.456; and 0.620/0.592 on `Claude-3-Opus`). Variance decomposition of Eq. (1) across all 284 OOF queries reveals the exact mechanism: `CodeBERT` patch distances occupy a narrow anisotropic cone ($d_{\text{patch}} = 0.0092 \pm 0.0051$, $\text{Var}(\frac{1}{2}d_{\text{patch}}) = 6.56 \times 10^{-6}$), whereas `MiniLM` log distances have 15.6× larger mean and 56.0× larger variance ($d_{\log} = 0.1433 \pm 0.0383$, $\text{Var}(\frac{1}{2}d_{\log}) = 3.67 \times 10^{-4}$). Thus, noisy log distances contribute **98.24**% of unweighted variance in $d(x, r)$, drowning out the patch channel.

Ablation 2: Reference contamination (H3). Table 4 and Figure 4(b) replace $r \in \{0, 1, 3\}$ of 32 normal references (0%, 3.1%, 9.4%) with training failures across 3 replacement seeds. **H3 is refuted:** neither `T2` ($|\Delta\text{BA}| \leq 0.003$, $p \geq 0.25$) nor `Inductive ECOD` ($|\Delta\text{BA}| \leq 0.003$, $p \geq 0.34$) degrades monotonically. In `R3`, contaminated references enter the k-NN set for 9.6%±2.0% (1/32) and 31.1% ± 7.6% (3/32) of test runs, but be-

Case ID	Type	Fold	St/Er/Ed	$d_{\log}/d_{\text{p}}$	$s(x)$	τ^* (Δ)
`django-13230`	FP	3 (16/32)	15/9/8	0.203/0.007	0.113	0.109 (+0.0041)
`sympy-12454`	FN	1 (5/32)	11/3/1	0.047/0.003	0.025	0.073 (-0.0483)
`sympy-24213`	FP*	1 (5/32)	16/11/1	0.144/0.003	0.073	0.073 (+0.0002)

Table 6: Traceable `T2` error cases (`FP*`: boundary FP) with fold (same-repo normal refs/32), raw **St**eps/**Er**rors/**Ed**its, $d_{\log}/d_{\text{patch}}$, score $s(x)$, and τ^* ($s - \tau^*$).

cause clean failure (0.0770 ± 0.0194) and normal (0.0732 ± 0.0200) distances already overlap heavily, mean failure scores shift by $< 0.01\,\text{SD}$ ($+0.00001$ and -0.00014). In `ECOD`, replacing $\leq 3/32$ ranks shifts smoothed tail log-probabilities by at most $\log(33/30) \approx 0.095$ nats/feature, which validation τ^* recalibration absorbs.

Ablation 3: Threshold calibration vs. operational cost model. Table 5 compares validation threshold rules against our stakeholder cost model ($C_{\text{Alert}} = \$30$ vs. $C_{\text{FN}} = \$1{,}200$). Enforcing $\text{FPR} \leq 20\%$ on 10 validation normals sets high cutoffs ($\bar{\tau}^* = 0.091$ for `T2`, 33.43 for `ECOD`), missing 179 and 128 failures ($381,602 and $276,180). Switching to validation F_2 / minimum-cost calibration lowers $\bar{\tau}^*$ to 0.046 and 17.05, raising recall to 0.957 ($F_2 = 0.924$) and 0.983 ($F_2 = 0.945$) and cutting cost by $> 89\%$ to $35,387 and $23,028. However, Bayes-optimal alerting flags whenever posterior failure probability exceeds $\pi^* = C_{\text{Alert}}/C_{\text{FN}} = 30/1200 = 2.5\%$. Because benchmark failure prevalence is $80.99\% \gg 2.5\%$, **Always alert** ($\tau = -\infty$, $15,000 vs. $365,400 unscreened) beats every automated filter. Anomaly detectors instead add distinct value on the rare 6.0% (17/284) **silent semantic failure** subtype, where `T1` catches $\mathbf{0/17}$ ($\mathbf{0.0}\%$), `ECOD` catches 2/17 (11.8%), `T2` catches 6/17 (35.3%), `T3` catches 7/17 (41.2%), and `Conditional Residual` catches $\mathbf{8/17}$ ($\mathbf{47.1}\%$).

8 Analysis

Traceable error cases and additive attribution. Table 6 and Figure 5(a) diagnose three deterministic `T2` errors: (1) **High-score FP `django-13230`** (Fold 3, $y = 0$, $s = 0.1128 > \tau^* = 0.1087$): despite strong repository coverage (16/32 normal `django` references in Fold 3), 15 steps with 9 error tracebacks inflate step-level log distance ($d_{\log} = 0.203$, Figure 5(a)), overriding its near-zero diff distance ($d_{\text{patch}} = 0.007$ to `django-11133`); (2) **Low-score FN `sympy-12454`** (Fold 1, $y = 1$, $s = 0.0248 < \tau^* = 0.0731$): a quiet 11-step, 1-error trajectory produces a clean 1-line edit ($d_{\log} = 0.047$, $d_{\text{patch}} = 0.003$ to `sympy-13647`) that misses an upper-triangular index bound, illustrating the silent semantic failure blind spot; and (3) **Boundary FP `sympy-24213`** (Fold 1, $y = 0$, $s = 0.0733 > \tau^* = 0.0731$, margin $+0.0002$): Fold 1 has only 5/32 normal `sympy` references and 1 normal `sympy` validation run, causing discrete quantile jumps in τ^* (leave-one-repository-out BA spans 0.25–1.00 across 10 estimable repos). Conversely, Figure 5(b) shows that `Inductive ECOD` yields exact additive feature attributions on TP `django-11630` ($s = 45.21 > \tau^* = 33.42$): four extreme patch tails (`2,310` chars, `64` lines, `3` files, edit ratio `1.00`) contribute 31.1% (7.8% each) and looping (`streak=15`, `stalled=14`) contributes 12.5%.

9 Conclusion and Limitations

Across 284 `GPT-4` trajectories and an untouched 271-run `Claude-3-Opus` holdout, frozen dual-encoder k-NN (`T2`, $\text{BA} = 0.481$) underperforms parameter-free `Inductive ECOD` ($\text{BA} = 0.657$, $p = 0.004$) and `Conditional Residual` ($\text{BA} = 0.675$, $p = 0.011$). Unweighted multimodal fusion fails because noisy `MiniLM` log distances carry 98.24% of channel variance, drowning out `CodeBERT` patch embeddings ($\text{BA} = 0.594$, $p = 0.012$). Under realistic release costs ($\pi^* = 2.5\%$), 81% failure prevalence favors universal review ($15,000), whereas anomaly detectors uniquely catch 47.1% of rare (6.0%) silent semantic failures missed by surface rules (0.0%). Limitations include small normal pools ($n_{\text{ref}} \in \{21, 32\}$), 10-normal validation quantization, orderless step pooling, and execution-free diff encoding; deployment requires recalibrating τ^* when rolling 50-run FPR exceeds 25% or when agent scaffolds/models upgrade.

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